Motivating Altruistic Behavior

W241 - Experiments and Causal Inference
Final Presentation

Sumedh Shah
Devashish Kulkarni
Ryan Wilson

Berkeley
UNIVERSITY OF CALIFORNIA

Research Purpose and Question

Examine altruistic behavior of survey participants by providing the experimental subjects with the specific benefits of persisting through the survey

The evolution of altruism suggest that psychological mechanisms, like emotions, promote altruistic behaviors, rather than conscious calculation of personal gain

Can we motivate altruistic behavior by outlining specific benefits for others?

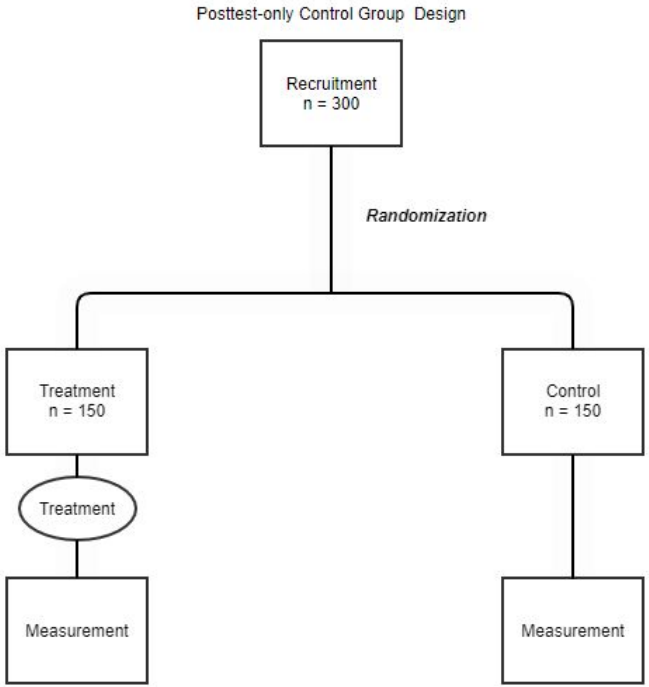
Hypothesis

Participants who are provided with altruistic survey instructions will answer more questions than participants who receive general survey instructions. We will control for sex, age, and ethnicity but specifically hypothesize that those of Eastern cultures will answer more questions than those of Western cultures because of their collectivistic culture.

Experimental Design

- **Subjects*:*** Participants recruited through Prolific platform
- **Randomization*:*** Randomization to treatment and control groups was completed by the survey platform Qualtrics.
- **Treatment*:*** Participants in the treatment group will be provided with information on how their responses will benefit development of a math app for underprivileged students
- **Outcome*:*** The average number of questions answered by the participants in the two groups and the average accuracy (correctly answered questions / number of questions answered)

Flow chart and ROXO comparison



Control Prompt

Thank you for participating in our survey. We are asking that you complete up to 5 puzzles in 4 minutes. If you correctly answer all 5 puzzles, you will be entered into a raffle to win a $50 Amazon gift card.

Treatment Prompt

Thank you for participating in our survey. We work in ed-tech and are developing a natural language processing math app for elementary and middle school mathematics. Your responses will help us train our algorithm to help provide real time feedback to students as they work on our platform. We are asking to complete up to 5 puzzles in 4 minutes. Answering all 5 puzzles will provide us a diverse set of responses to provide meaningful feedback to the students using our platform. If you answer all 5 puzzles correctly, you will also be entered into a raffle to win a $50 Amazon gift card.

Puzzles: Factor 24

Q25

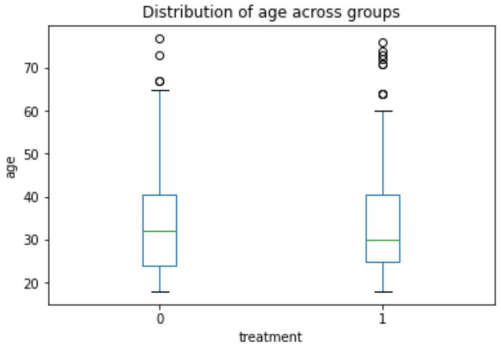
Example:

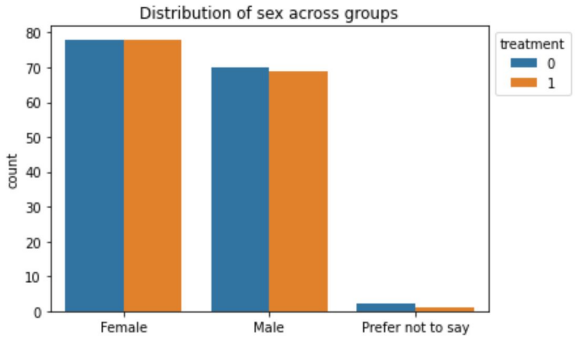
Factors: 6, 3, 2, 3
Step 1: 6 * 3 = 18
Step 2: 2 * 3 = 6
Step 3: 18 + 6 = 24

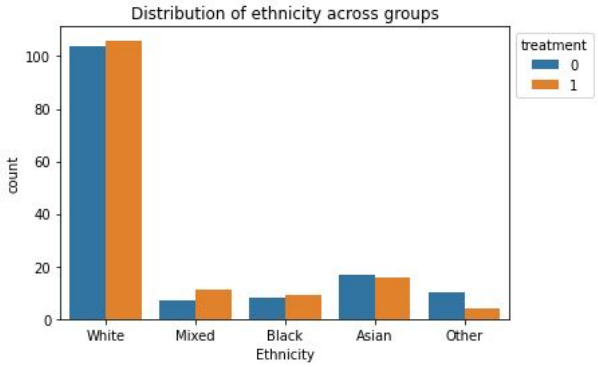
Enter **6*3 + 2*3** as your response for the puzzle.

Please note that some of the puzzles may have multiple answers.

Covariate Balance Check

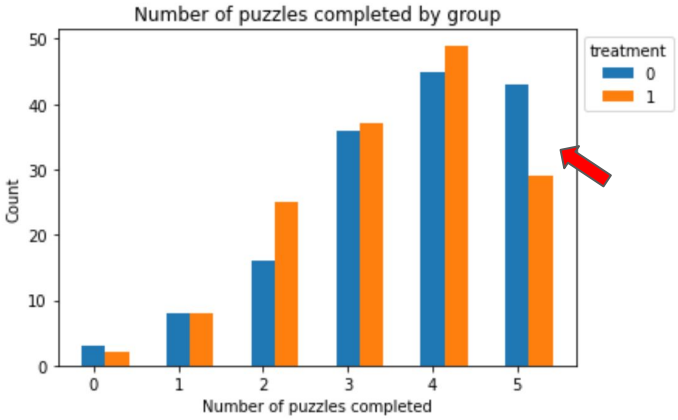


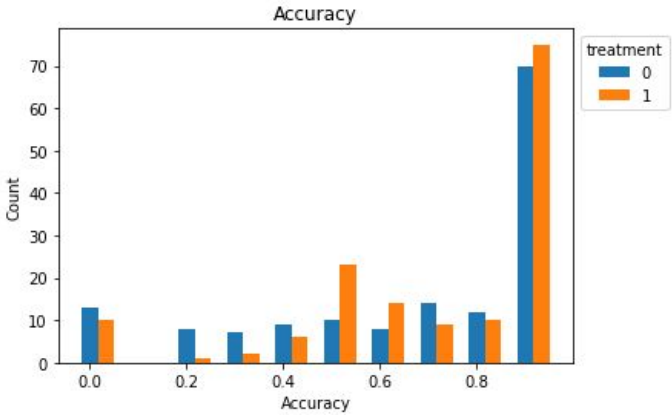




- Similar distribution of covariates across two groups - Randomization seems to have worked well.
- Sample size: 300

Outcome distribution





- More participants in control attempted all 5 questions, compared to treatment (43 vs 29)
- chisq.test(x=c(43,29), p =c(.5,.5)) –> p-value of **0.09896**
- **Hypothesis**: Control group more likely to attempt all 5 puzzles for the raffle, Treatment group was careful to get the puzzles right and took longer for the first 4 puzzles.

- Similar distribution of accuracy across groups
- Treatment did not affect participant performance

Experiment Results and Analysis

	Dependent variable:		
	num_completed		
	(1)	(2)	(3)
treatment	-0.196	-0.202	-0.169
	(0.143)	(0.145)	(0.156)
ethnicityAsian		-0.086	0.057
		(0.205)	(0.280)
sexMale		0.204	0.205
		(0.145)	(0.145)
sexNaN		-0.916	-0.915
		(2.084)	(2.108)
sexPrefer not to say		0.197	0.203
		(0.446)	(0.439)
age		-0.004	-0.004
		(0.006)	(0.006)
treatment:ethnicityAsian			-0.294
			(0.404)
Constant	3.596***	3.665***	3.647***
	(0.103)	(0.230)	(0.231)
Observations	301	298	298
R2	0.006	0.020	0.022
Adjusted R2	0.003	0.00001	-0.002
Residual Std. Error	1.233 (df = 299)	1.226 (df = 291)	1.227 (df = 290)
F Statistic	1.903 (df = 1; 299)	1.001 (df = 6; 291)	0.916 (df = 7; 290)
Note:			*p<0.1; **p<0.05; ***p<0.01

- **ATE is -0.196** for those in treatment group with **average of 3.596 questions answered as base**
- No statistically significant effect of treatment observed
- Covariates don't really affect outcome and not significantly different from 0
- Inclusion of Asian ethnicity covariate & interaction with treatment barely affected outcome

Experiment Results and Analysis

	Dependent variable:		
	accuracy		
	(1)	(2)	(3)
treatment	0.043	0.043	0.050
	(0.036)	(0.037)	(0.040)
ethnicityAsian		0.088*	0.118*
		(0.052)	(0.069)
sexMale		-0.007	-0.006
		(0.038)	(0.038)
sexNaN		-0.108	-0.108
		(0.470)	(0.466)
sexPrefer not to say		0.181*	0.182*
		(0.098)	(0.098)
age		-0.002	-0.002
		(0.002)	(0.002)
treatment:ethnicityAsian			-0.061
			(0.105)
Constant	0.719***	0.781***	0.778***
	(0.027)	(0.063)	(0.064)
Observations	301	298	298
R2	0.005	0.026	0.027
Adjusted R2	0.001	0.006	0.004
Residual Std. Error	0.315 (df = 299)	0.316 (df = 291)	0.316 (df = 290)
F Statistic	1.423 (df = 1; 299)	1.306 (df = 6; 291)	1.156 (df = 7; 290)
Note:			*p<0.1; **p<0.05; ***p<0.01

- **ATE is 0.043 (4.3% accuracy)** with **average accuracy of 0.719 (71.9%) as base**
- No statistically significant effect of treatment observed
- Statistically significant effect (p<0.1) of Asian ethnicity on accuracy (8.8% and 11.8% more accuracy for second two regressions)
- Inclusion of Asian ethnicity covariate and interaction with treatment barely affected outcome

Conclusion

- Minimal change detected in number of puzzles completed and accuracy between control and treatment
  - Addition of covariates and HTE barely made any difference
- People might've completed less puzzles in treatment in order to give more accurate answers
- For proper treatment, might need to administer experiment away from Prolific (in person) to random friends, family members, and coworkers
  - Prolific users might be clicking through and not properly reading instructions
- Need to find sweet spot in time limit for puzzles